

Cycles 1-2 Clone 2 Report — M-RECREATE-1/second-real-audio-batch (Fork 33a2a8003c84)

2026-08-29

Contents

Cycles 1-2 Clone 2 Report — M-RECREATE-1/second-real-audio-batch (Fork 33a2a8003c84)	1
Abstract	1
Verdict	1
Rubric SHA Anchor Chain (Three-Way Byte-Equal)	2
5-Song Selection (SHA-256 Tiebreak Per Bucket; c37 Clone-0 Song Excluded)	2
Batch Verdict Numerics	2
Cross-Band Correlations (n=5 Sample-Size Caveat Literal on Every Row)	2
Byte-Determinism × 2 (20/20 SHA Pairs Matched)	3
Anchor Preservation (18/18 c37 Read-Only Anchors Byte-Identical)	3
preview_untrained_ear Caveat (v1-Branch Conditional Handling)	3
Test Surface	3
Cycle Disposition	3
State-Machine Discipline (c29 Lemma Respected)	3
Ledger Events (10 Shadow Rows Under -clone-2 Suffix; Cycle 2: 0)	4
Merge Disposition (Root-Conductor Post-Merge Validation Checklist)	4
Standing Constraints (Unchanged)	4
Anti-Patterns Locked (5-Count Stable)	4
Effects-Chain Band-Selectivity Signal (Deferred to c40+)	5
Cycle-39 Handoff (Forward-Looking, Owned by c39 Fanout Not This Branch)	5
Cumulative Progress	5

[OUTPUT: report_cycles_1-2_clone_2]

Cycles 1-2 Clone 2 Report — M-RECREATE-1/second-real-audio-batch (Fork 33a2a8003c84)

Abstract

Cycles 1-2 of clone-2 (fork 33a2a8003c84) close M-RECREATE-1/second-real-audio-batch at **BATCH_LANDS**. Cycle 1 executed the 5-song batch pipeline via SHA-256 tiebreak per rating bucket (excluding the c37 clone-0 song), consuming c37 scripts/recreate_v0/* machinery read-only, using the Stage-06 pretty_midi fallback as-is, and producing positive mel_l1_db effects deltas on all 5 songs (+2.879 to +7.983 dB; mean +5.04 dB). Cycle 2 is c30-codified verification-only standby (zero writes; auditor decision **COMPLETE**; [[BRANCH_COMPLETE]] emitted). 15/16 in-clone rubric criteria PASS; the 16th (git-log gate) is legitimately MERGE_DEFERRED to the root conductor per known clone-environment git add refusal.

Verdict

BATCH_LANDS (VALIDATED at cycle 1; **COMPLETE** at cycle 2 standby; [[BRANCH_COMPLETE]]).

Rubric SHA Anchor Chain (Three-Way Byte-Equal)

Location	SHA-256
docs/recreate_v0_batch_rubric.md	be65f7cb37f71c4613afceb70dafa03a1bc68a384f4b51639127b4d5256b7
data/recreate_v0_batch/rubric_hash.txt	be65f7cb...b718d
verdict.json.rubric_hash	be65f7cb...b718d

Rubric mtime 08:50 < earliest scripts/recreate_v0_batch/*.py mtime 08:51+. git-log gate **MERGE_DEFERRED** (clone environment refuses git add/git commit; worker honestly recorded rather than fabricating; conductor verifies rubric-first commit order post-integration per c38 precedent).

5-Song Selection (SHA-256 Tiebreak Per Bucket; c37 Clone-0 Song Excluded)

Songs selected via SHA-256 tiebreak from each of the four rating buckets (band-4, band-5, band-6, band-7) with an extra tiebreak within the largest bucket, excluding corpus/ratings/7/016__LOCAL__05_02.mp3 (c37 clone-0 song). All 5 songs complete the 8-stage c37 pipeline (decode → chunker → tagger sidecar → htdemucs → basic-pitch × 3 → score merge → fluidsynth bare-MIDI → cycle-9 DawDreamer effects).

Batch Verdict Numerics

Metric	Result
n_pipeline_ok	5/5
n_byte_det_x2	5/5
n_positive_mel_delta	5/5 (per-song mel deltas +2.879 to +7.983 dB; mean +5.04)
Cross-band table dimensions	5 rows × 14 numeric cols

Cross-band table includes mel_l1_db + spectral_centroid_rmse_hz + rms_env_rmse + lufs_m_rmse per the brief, plus n=5 exploratory caveat literal on every correlation row.

Cross-Band Correlations (n=5 Sample-Size Caveat Literal on Every Row)

On-disk data/recreate_v0_batch/cross_band_correlation.json and docs/recreate_v0_batch_report.md (correct values):

Metric	r (band vs improvement)
mel_l1_db	−0.483
spectral_centroid_rmse_hz	+0.911
rms_env_rmse	−0.199
lufs_m_rmse	−0.695

MINOR-tier drift note: worker's prior compacted-session narrative-to-auditor text cited stale correlation numbers (mel −0.180, spec +0.858, rms −0.535, lufs −0.826); on-disk artefacts

show the correct values above. All on-disk consumers internally coherent; narrative-vs-artefact drift only, not a scientific defect.

Byte-Determinism × 2 (20/20 SHA Pairs Matched)

5 songs × 4 anchors (merged.musicxml, merged.midi, bare_midi.wav, effects.wav) = 20 anchor SHAs; all byte-equal across two independent runs (determinism_check.json.all_equal = true).

Anchor Preservation (18/18 c37 Read-Only Anchors Byte-Identical)

anchor_preservation.json: 18 anchors under scripts/recreate_v0/*, unchanged: true, changed: {}. Zero writes under c37 clone-0's machinery; read-only import via PYTHONPATH per rubric §7.

preview_untrained_ear Caveat (v1-Branch Conditional Handling)

Branch A's v1 report landed mid-execution at 09:07; clone-2 report cites docs/ear_real_label_training_v1_report.md **by document path only**, never programmatically importing v1 model, verdict.json, or corn_head_v1.pt. Falls back to v0 caveat citation (M-EAR-1/real-label-training-v0 → EAR_v0_INSUFFICIENT per c36) if the v1 report path is not present. Pattern crystallised for reuse across c39 HS-2 and future recreation cycles.

Test Surface

Suite	Result
tests/test_recreate_v0_batch.py	15/15 PASS (exceeds ≥12 minimum)
python3 -m long_exposure.tools.promise_check .	0 ERRORS (pre-existing WARNs only; no new WARNs)
org_check	0 ERRORS (no new WARNs introduced)

Cycle Disposition

Cycle	Researcher Directive	Worker Action	Auditor Decision
1	Ship the milestone under frozen 3-verdict rubric	5-song batch pipeline; BATCH_LANDS with 5/5 positive mel deltas; 10 shadow-ledger rows	VALIDATED
2	Verification-only standby	Honest low-output; refused to fabricate activity or adopt forward-looking c39 scope	COMPLETE ([[BRANCH_COMPLETE]])

State-Machine Discipline (c29 Lemma Respected)

M-RECREATE-1/second-real-audio-batch is a peer sub-milestone under M-RECREATE-1. NOT a child of terminal-validated c37 clone-0 M-RECREATE-1/first-real-audio. c29 lemma preserved; ledger topology stays a DAG.

Ledger Events (10 Shadow Rows Under -clone-2 Suffix; Cycle 2: 0)

Six substantive M-RECREATE-1/second-real-audio-batch/* events auto-suffixed -clone-2 + 4 house-keeping under -clone-2 per c33 writer-guard behavior:

1. _run/cycle_38_launched-clone-2 (status: validated per c35 Branch C codified convention)
2. _plan/recreate_v0_batch_rubric_frozen-clone-2
3. _infra/egress-probe-cycle-38-clone-2
4. M-RECREATE-1/second-real-audio-batch-clone-2 (in-progress; auto-suffixed by c33 writer-guard)
5. M-RECREATE-1/second-real-audio-batch/song-selection-clone-2
6. M-RECREATE-1/second-real-audio-batch/pipeline-executed-clone-2
7. M-RECREATE-1/second-real-audio-batch/byte-determinism-verified-clone-2
8. M-RECREATE-1/second-real-audio-batch-clone-2 (validated verdict roll-up, BATCH_LANDS)
9. _run/cycle_38_closed-clone-2
10. _archive/cycle-38-scratch-clone-2 + _infra/adopt-cycle38-tests-clone-2 (housekeeping pair)

Cycle 2: zero. validated → in_progress forbidden per c29 lesson.

Merge Disposition (Root-Conductor Post-Merge Validation Checklist)

Merge report at /home/user/music-gen-instance/fork-33a2a8003c84/clone-2/merge_report.md for root conductor pickup.

1. Read 10-row shadow ledger; reconcile with clone-0's ledger for the same fork.
2. git add + git commit on-disk artefacts in rubric-first order (docs/recreate_v0_batch_rubric.md first, then scripts/recreate_v0_batch/*, then data/recreate_v0_batch/*, then docs/recreate_v0_batch_report.md, then tests/test_recreate_v0_batch.py); validate git-log gate post-commit.
3. Run promise_check; expect 0 ERRORS.
4. Verify _manager/fanout-namespace-convention-discrepancy open ticket (c33 writer-guard auto-suffixes substantive M-* labels contrary to c32 convention doc); resolution deferred to c40+ (narrow guard OR update doc).
5. Write merge_report.md at the fork-clone path for root conductor pickup.

Standing Constraints (Unchanged)

- α pinned at 0.7469387071101908 (not relevant to this branch).
- SHA-256 tiebreak; no PRNG; no sidecar_nonfactor.
- Interpreter guard #!/usr/bin/python3 on every new script.
- Read-only anchors preserved: 18/18 under scripts/recreate_v0/* (c37 clone-0 machinery), c9 DawDreamer effects chain, c22 stability harness, c26 Path B commitment, c31/c33/c34/c35/c36 palette anchors.
- Rated audio egress-blocked at *.googlevideo.com (unchanged 403; retry cadence at conductor level; not required — 5 songs are on-disk operator-delivered corpus).
- Ledger hygiene: narrative field; run_id="run-2026-08-28T040704Z"; nested confidence:{level,rationale,asses
UUID5 content-hash event_id; two-arg append_ledger_event(workspace, event).

Anti-Patterns Locked (5-Count Stable)

c8 octave-suppression; c11 CLAP/VGGish embedding; c22 stability; c23 head-reg; c25 feature-representation — not re-attempted. c31 STILL_GAP / c35 A anti-pattern surface intact. c30 collision-arc closure at PARTIAL_BP_UNRESOLVED_SHAPE unchanged.

Cycle-2 discipline correctly avoided the “gold plate” anti-pattern by refusing to re-do validated work, refusing to adopt forward-looking c39 scope not assigned to this clone, and refusing to fabricate an activity log where none was warranted.

Effects-Chain Band-Selectivity Signal (Deferred to c40+)

Per-song mel deltas span $2.7\times$ (+2.88 to +7.98 dB across 5 songs), suggesting the c9 pinned effects chain is not equally-well-tuned across musical-content classes. Queued as `_manager/effects-chain-band-selectivity` for c40+ opportunistic pickup, or promoted to urgent if c39 HS-2 (n=37) shows per-band mel-delta failures.

Cycle-39 Handoff (Forward-Looking, Owned by c39 Fanout Not This Branch)

Per research brief:

1. **HS-1 M-EAR-1/real-label-training-v1/out-of-fold-cross-band** — analytical study of the SB1/SB2 partial-corpus failures under out-of-fold band evaluation.
2. **HS-2 M-RECREATE-1/full-corpus-recreation** — extend to n=37 (or 42 pooled with c38 clone-2 batch) to establish first full-corpus real-audio recreation measurement with cross-band effects-delta trends. G1 spine promotion: c37 (n=1) → c38 (n=5) → c39 (n=37/42).

c39 workers accept whatever the writer emits re namespace convention; merge conductor to resolve `_manager/fanout-namespace-convention-discrepancy` durably at c40+.

Cumulative Progress

M-RECREATE-1 arc:

Cycle	Milestone	Verdict	Songs
c37 clone-0	M-RECREATE-1/first-real-audio	RECREATION_ LANDS	n=1 (band-7 song 016)
c38 clone-2 (this)	M-RECREATE-1/second-real-audio-batch	BATCH_LANDS	n=5 (SHA-256 tiebreak per bucket)
c39 HS-2 (forward)	M-RECREATE-1/full-corpus-recreation	(target)	n=37 (or 42 pooled)

G1 spine promotion is on track: c37 (n=1) → c38 (n=5) → c39 HS-2 (n=37 target) will establish the first full-corpus real-audio recreation measurement, complete with cross-band effects-delta trends.

Pre-registration discipline holds across three consecutive real-audio recreation cycles: frozen rubric-first commit order + mtime-gate + git-log-gate (with `MERGE_DEFERRED` clone-env accommodation) has become the campaign’s stable pattern for adjudication-heavy work.

preview_untrained_ear caveat handling crystallised: v1-branch cites Branch A v1 report by document path only, never imports v1 model or verdict.json programmatically; v0-branch cites c36 M-EAR-1/real-label-training-v0 INSUFFICIENT verdict. Pattern reusable across c39 HS-2 and future recreation cycles.

Verification-only cycle discipline reinforced at cycle 2: worker’s honest “no new work performed this session” self-report is correct professional conduct at cycle tail. `[[BRANCH_COMPLETE]]` emitted per auditor role definition (milestone validated AND scope fully discharged).

Namespace convention discrepancy (c33 writer-guard auto-suffixes substantive M-* while c32 convention doc says unsuffixed) remains open as _manager/fanout-namespace-convention-discrepancy; merge conductor to resolve durably at c40+.

c29 state-machine lemma respected: peer sub-milestone; ledger topology stays a DAG. **G5 collision-modeling arc**: closed at PARTIAL_BP_UNRESOLVED_SHAPE (c30 terminal); no re-opening proposed.

[END OUTPUT]